

Qatar-2b

R-0924

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_180529 · created 2026-07-29T09:45:19+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458267.74659 +/- 0.00063 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.001 +/- 0.066
Transit depth (Rp/Rs)^2	0.00014 +/- 0.015 [%]
Orbital Inclination (inc)	85.98 +/- 1.74
Airmass coefficient 1 (a1)	1.92 +/- 1.14
Airmass coefficient 2 (a2)	-0.45 +/- 0.39
Scatter in the residuals of the lightcurve fit is	58.8 %
Best Comparison Star	None
Optimal Aperture	1.34
Optimal Annulus	3.25
Transit Duration (day)	0.0616 +/- 0.0077

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.001 ± 0.066 vs 0.16327 (deviation 2.46σ)
Duration fit vs published	0.0616 d vs 0.0767 d (deviation 1.96σ)
Mid-time O–C	0.1 min
Depth SNR	0.0
Residual scatter	58.8 %

Light curve

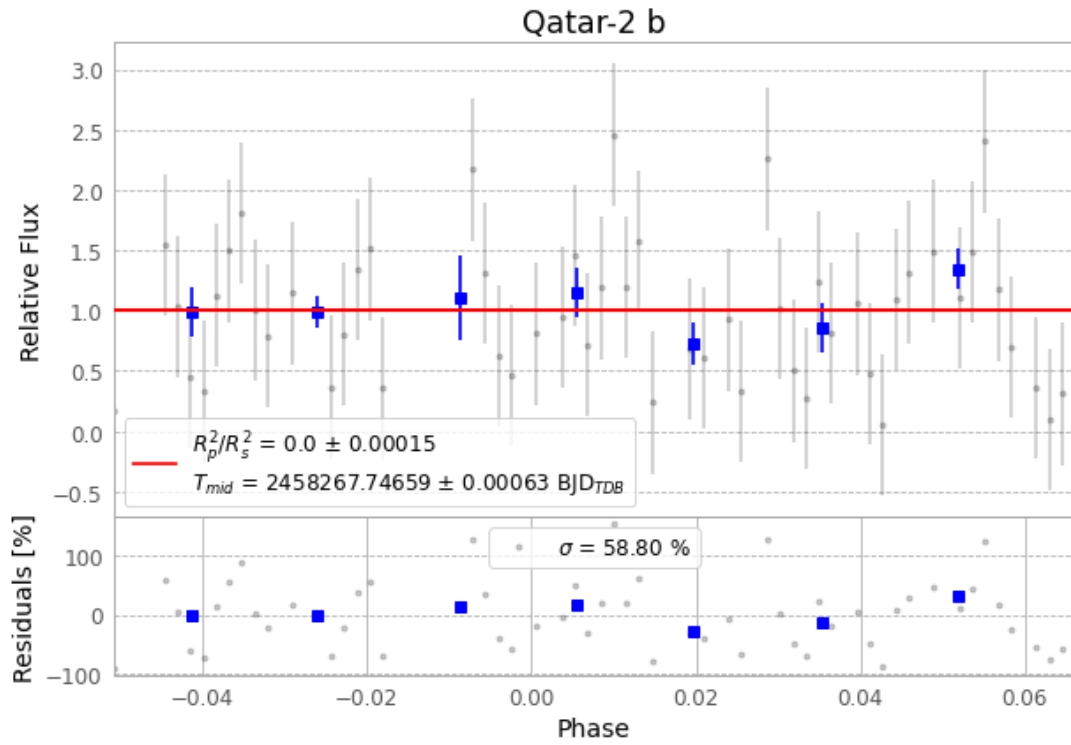


Figure 1 — FinalLightCurve_Qatar-2 b_2018-05-28.png (SHA-256 372f3fa4fbe91b71...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [392, 423], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 53cff088b1334924...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

76 FITS frames (50 MB), Qatar-2180529040912.FITS → Qatar-2180529075412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-180529.log (a4d80610669d...), FinalLightCurve_Qatar-2 b_2018-05-28.png (372f3fa4fbe9...), FinalLightCurve_Qatar-2 b_2018-05-28.csv (1dbdce57076f...)

Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 09:45Z.